

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

- If $x \log_e(\log_e x) - x^2 + y^2 = 4 (y > 0)$, then dy/dx at $x = e$ is equal to :

(A) $\frac{e}{\sqrt{4+e^2}}$ (B) $\frac{(1+2e)}{2\sqrt{4+e^2}}$ (C) $\frac{(2e-1)}{2\sqrt{4+e^2}}$ (D) $\frac{(1+2e)}{\sqrt{4+e^2}}$

MDF016
- If $y = \ell n \left\{ \frac{x + \sqrt{a^2 + x^2}}{a} \right\}$, then the value of $\frac{dy}{dx}$ is-

(A) $\sqrt{a^2 - x^2}$ (B) $a \sqrt{a^2 + x^2}$ (C) $\frac{1}{\sqrt{a^2 + x^2}}$ (D) $x \sqrt{a^2 + x^2}$

MDF058
- Let $u(x)$ and $v(x)$ are differentiable functions such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{u(x)}{v(x)} \right)' = q$, then $\frac{p+q}{p-q}$ has the value equal to -

(A) 1 (B) 0 (C) 7 (D) -7

MDF059
- If $y = \sqrt{\frac{1-x}{1+x}}$, then $\frac{dy}{dx}$ equals-

(A) $\frac{y}{1-x^2}$ (B) $\frac{y}{x^2-1}$ (C) $\frac{y}{1+x^2}$ (D) $\frac{y}{y^2-1}$

MDF001
- If $x = y \ell n(xy)$, then $\frac{dx}{dy}$ equals-

(A) $\frac{y(x-y)}{x(x+y)}$ (B) $\frac{x(x+y)}{y(x-y)}$ (C) $\frac{y(x+y)}{x(x-y)}$ (D) $\frac{x(x-y)}{y(x+y)}$

MDF002
- If $(\cos x)^y = (\sin y)^x$, then $\frac{dy}{dx}$ equals -

(A) $\frac{\log \sin y - y \tan x}{\log \cos x + x \cot y}$ (B) $\frac{\log \sin y + y \tan x}{\log \cos x - x \cot y}$ (C) $\frac{\log \sin y + y \tan x}{\log \cos x + x \cot y}$ (D) $\frac{\log \sin y + y \tan x}{\log \cos y - y \cot x}$

MDF003
- $\frac{d}{dx} \left(\tan^{-1} \left(\frac{\sqrt{x-x}}{1+x^{3/2}} \right) \right)$ equals- ($x \geq 0$)

(A) $\frac{1}{2\sqrt{x}(1+x)} - \frac{1}{1+x^2}$ (B) $\frac{1}{2\sqrt{x}(1+x)} + \frac{1}{1+x^2}$ (C) $\frac{1}{1+x} - \frac{1}{1+x^2}$ (D) $\frac{1}{1+x} + \frac{1}{1+x^2}$

MDF005

8. Let f be a function defined for all $x \in R$. If f is differentiable and $f(x^3) = x^5$ for all $x \in R$ ($x \neq 0$), then the value of $f'(27)$ is-
- (A) 15 (B) 45 (C) 0 (D) 35

MDF007

9. If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$ are given, then -

- (A) $\Delta_1 = 3(\Delta_2)^2$ (B) $\frac{d}{dx} \Delta_1 = 3\Delta_2$ (C) $\frac{d}{dx} \Delta_1 = 3(\Delta_2)^2$ (D) $\Delta_1 = 3(\Delta_2)^{3/2}$

MDF008

10. Let $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in A.P., then $f'(a), f'(b)$ and $f'(c)$ are in
- (A) G.P. (B) H.P. (C) A.G.P. (D) A.P.

MDF009

11. Suppose the function $f(x) - f(2x)$ has the derivative 5 at $x = 1$ and derivative 7 at $x = 2$. The derivative of the function $f(x) - f(4x)$ at $x = 1$, has the value equal to
- (A) 19 (B) 9 (C) 17 (D) 14

MDF010

12. If $f(x) = \begin{cases} x \tan^{-1} x + \sec^{-1}(1/x), & x \in (-1, 1) - \{0\} \\ \pi/2, & \text{if } x = 0 \end{cases}$, then $f'(0)$ is -

- (A) equal to -1 (B) equal to 0 (C) equal to 1 (D) non existent

MDF011

13. If $x = t^3 + t + 5$ & $y = \sin t$, then $\frac{d^2y}{dx^2} =$

- (A) $-\frac{(3t^2+1)\sin t + 6t \cos t}{(3t^2+1)^3}$ (B) $\frac{(3t^2+1)\sin t + 6t \cos t}{(3t^2+1)^2}$
(C) $-\frac{(3t^2+1)\sin t + 6t \cos t}{(3t^2+1)^2}$ (D) $\frac{\cos t}{3t^2+1}$

MDF012

14. Let $f(x) = x + \sin x$. Suppose g denotes the inverse function of f . The value of $g'\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right)$ has the value equal to

- (A) $\sqrt{2}-1$ (B) $\frac{\sqrt{2}+1}{\sqrt{2}}$ (C) $2-\sqrt{2}$ (D) $\sqrt{2}+1$

MDF015

15. If $y = x + e^x$ then $\frac{d^2x}{dy^2}$ is :

- (A) e^x (B) $-\frac{e^x}{(1+e^x)^3}$ (C) $-\frac{e^x}{(1+e^x)^2}$ (D) $\frac{-1}{(1+e^x)^3}$

MDF013

16. If $2y = \left(\cot^{-1} \left(\frac{\sqrt{3} \cos x + \sin x}{\cos x - \sqrt{3} \sin x} \right) \right)^2$, $x \in \left(0, \frac{\pi}{2} \right)$, then $\frac{dy}{dx}$ is equal to :

- (A) $2x - \frac{\pi}{3}$ (B) $\frac{\pi}{3} - x$ (C) $\frac{\pi}{6} - x$ (D) $x - \frac{\pi}{6}$

MDF017

17. If $f(1) = 1$, $f'(1) = 3$, then the derivative of $f(f(f(x))) + (f(x))^2$ at $x = 1$ is :

- (A) 12 (B) 33 (C) 9 (D) 15

MDF018

18. Suppose that $f(0) = 0$ and $f'(0) = 2$, and let $g(x) = f(-x + f(f(x)))$. The value of $g'(0)$ is equal to

- (A) 0 (B) 1 (C) 6 (D) 8

MDF060

19. Let $f : (-1, 1) \rightarrow R$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let $g(x) = [f(2f(x) + 2)]^2$. Then $g'(0) :-$

- (A) 4 (B) -4 (C) 0 (D) -2

MDF020

20. If $f(4) = g(4) = 2$; $f'(4) = 9$; $g'(4) = 6$ then $\lim_{x \rightarrow 4} \frac{\sqrt{f(x)} - \sqrt{g(x)}}{\sqrt{x} - 2}$ is equal to

- (A) $3\sqrt{2}$ (B) $\frac{3}{\sqrt{2}}$ (C) 0 (D) none of these.

MDF019

21. Let $e^{f(x)} = \ell nx$. If $g(x)$ is the inverse function of $f(x)$, then $g'(x)$ equals to:

- (A) e^x (B) $e^x + x$ (C) e^{x+e^x} (D) $e^{x+\ell nx}$

MDF056

22. If $8f(x) + 6f\left(\frac{1}{x}\right) = x + 5$ and $y = x^2 f(x)$, then $\frac{dy}{dx}$ at $x = -1$ is equal to

- (A) 0 (B) $\frac{1}{14}$ (C) $-\frac{1}{14}$ (D) None of these

MDF057

23. If $\frac{d^2x}{dy^2} \left(\frac{dy}{dx} \right)^3 + \frac{d^2y}{dx^2} = K$, then the value of K is equal to

- (A) 1 (B) -1 (C) 2 (D) 0

MDF061

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If $f(x) = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x$ and g is the inverse function of f , then
- (A) $g'(2\pi) = \frac{7}{3}$ (B) $g'(2\pi) = \frac{3}{7}$ (C) $g''(2\pi) = \frac{7}{3}$ (D) $g''(2\pi) = 0$

MDF062

2. Let $f(x) = \frac{\sqrt{x - 2\sqrt{x - 1}}}{\sqrt{x - 1} - 1} \cdot x$, then
- (A) $f'(10) = 1$ (B) $f'(3/2) = -1$
 (C) domain of $f(x)$ is $x \geq 1$ (D) none

MDF063

3. If $y = x^{(\ell n x)^{\ell n(\ell n x)}}$, then $\frac{dy}{dx}$ is equal to :
- (A) $\frac{y}{x} (\ell n x^{\ell n x - 1} + 2 \ell n x \ell n(\ell n x))$ (B) $\frac{y}{x} (\ell n x)^{\ell n(\ell n x)} (2 \ell n(\ell n x) + 1)$
 (C) $\frac{y}{x \ell n x} ((\ell n x)^2 + 2 \ell n(\ell n x))$ (D) $\frac{y \ell n y}{x \ell n x} (2 \ell n(\ell n x) + 1)$

MDF064

4. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is equal to
- (A) $-\frac{2^y}{2^x}$ (B) $\frac{1}{1 - 2^x}$ (C) $1 - 2^y$ (D) $\frac{2^x(1 - 2^y)}{2^y(2^x - 1)}$

MDF065

5. If $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$, then
- (A) $f(-2) = 0$ (B) $f'(-1/2) = 0$ (C) $f'(-1) = -2$ (D) $f''(0) = 4$

MDF066

6. The functions $u = e^x \sin x$; $v = e^x \cos x$ satisfy the equation
- (A) $v \frac{du}{dx} - u \frac{dv}{dx} = u^2 + v^2$ (B) $\frac{d^2u}{dx^2} = 2v$
 (C) $\frac{d^2v}{dx^2} = -2u$ (D) $\frac{du}{dx} + \frac{dv}{dx} = 2v$

MDF067

7. The differential coefficient of $\sin^{-1} \frac{t}{\sqrt{1+t^2}}$ w.r.t. $\cos^{-1} \frac{1}{\sqrt{1+t^2}}$ is
- (A) $1 \forall t > 0$ (B) $-1 \forall t < 0$ (C) $1 \forall t \in \mathbb{R}$ (D) none of these

MDF081

8. If $f(\theta) = \tan\left(\sin^{-1}\sqrt{\frac{2}{3+\cos 2\theta}}\right)$, then-

(A) $f'\left(\frac{\pi}{4}\right) = \sqrt{2}$

(B) $\frac{d(f(\theta))}{d(\cos\theta)} = -2$ at $\theta = \frac{\pi}{4}$

(C) $\frac{d(f(\theta))}{d(\cos\theta)} = -\frac{1}{2}$ at $\theta = \frac{\pi}{4}$

(D) $\frac{d(f(\theta))}{d(\sec\theta)} = 1$ at $\theta = \frac{\pi}{3}$

MDF082

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 9 to 11

Let $a_1 > a_2 > a_3 \dots a_n > 1$; $p_1 > p_2 > p_3 \dots > p_n > 0$; such that $p_1 + p_2 + p_3 + \dots + p_n = 1$.

Also $F(x) = (p_1 a_1^x + p_2 a_2^x + \dots + p_n a_n^x)^{1/x}$ then

9. $\lim_{x \rightarrow 0^+} F(x)$ is

(A) $a_1^{p_1} \cdot a_2^{p_2} \dots a_n^{p_n}$

(B) a_1

(C) a_n

(D) does not exist

MDF083

10. $\lim_{x \rightarrow \infty} F(x)$

(A) $a_1^{p_1} \cdot a_2^{p_2} \dots a_n^{p_n}$

(B) a_1

(C) a_n

(D) does not exist

MDF084

11. $\lim_{x \rightarrow -\infty} F(x)$

(A) $a_1^{p_1} \cdot a_2^{p_2} \dots a_n^{p_n}$

(B) a_1

(C) a_n

(D) does not exist

MDF085

Paragraph for Question 12 to 14

Consider the function defined implicitly $y^2 + y - x = 2$ on various intervals on the real line. If

$y \in \left(-\infty, -\frac{1}{2}\right)$, then equation implicitly defines a unique real valued differentiable function

$y = f(x)$. On the basis of above information's, answer the following questions.

12. The value of $f''(4)$ is -

(A) $\frac{2}{125}$

(B) $\frac{125}{2}$

(C) $\frac{1}{5}$

(D) $\frac{1}{125}$

MDF086

13. If $y = g(x)$ is inverse of $y = f(x)$, then $g'(-2)$ is -

(A) $-\frac{1}{3}$

(B) -3

(C) -2

(D) 0

MDF087

14. If $y = h(x)$ is mirror image of $y = f(x)$ about the line $2y + 1 = 0$, then $f''(x_0) + h''(x_0)$, where

$x_0 \in \left(-\frac{9}{4}, \infty\right)$ is -

(A) $2x_0$

(B) $\frac{2}{2f(x_0)+1}$

(C) $\frac{2}{(2f(x_0)+1)^3}$

(D) 0

MDF088

MATCHING LIST TYPE QUESTION

15. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II
(P) Let $f(x) = x^2 - 4x - 3, x > 2$ and let g be the inverse of f . Find the value of g' at $x = 2$.	(1)	-1
(Q) Let $f: R \rightarrow R$ be defined as $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$ and $g(x) = f^{-1}(x)$, then find $g'(-1)$.	(2)	$\frac{1}{6}$
(R) Suppose f^{-1} is the inverse function of a differentiable function f and let $G(x) = \frac{1}{f^{-1}(x)}$. If $f(3) = 2$ and $f'(3) = \frac{1}{9}$, find $G'(2)$.	(3)	1
(S) If $y = y(x)$ and it follows the relation $e^{xy} + y \cos x = 2$, then $y'(0) + y''(0)$.	(4)	$\frac{1}{14}$
(A) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$	(B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$	
(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$	(D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$	

MDF089

16. Match the following List-I with List-II

List-I		List-II
(P) Let $f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = x^2 + xf'(2) + f''(3)$, then $f'(1) - f'(2)$ is equal to	(1)	2
(Q) If $f(x - y), f(x), f(y)$ and $f(x + y)$ are in A.P., for all x, y and $f(0) \neq 0$, then $f'(-2) + f'(2)$ is equal to	(2)	1
(R) If $f(x) = x^3 + x^2, f'(1) + xf''(2) + f'''(3)$ for all, then $f(0) + f(3) + 1$ is equal to	(3)	4
(S) Let $f(x) = x^n, n$ being a positive integer, then value of n for which the equality $f'(a + b) = f'(a) + f'(b)$ is valid for all $a, b > 0$ is	(4)	0
(A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$	(B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$	
(C) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$	(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$	

MDF090

EXERCISE (S)

1. At $x = \frac{\pi^2}{4}$, $\frac{d}{dx} \cos(\sin\sqrt{x}) =$ MDF024
2. If $y\sqrt{x^2+1} = \log\{\sqrt{x^2+1} - x\}$, then $(x^2+1)\frac{dy}{dx} + xy + 1 =$ MDF027
3. If $f(x) = x + 2$, then $f'(f(x))$ at $x = 4$ is MDF028
4. Let $3f(x) - 2f(1/x) = x$, then $4f'(2)$ is equal to MDF029
5. If $x = 2\cos t - \cos 2t$ & $y = 2\sin t - \sin 2t$, find the value of $2\left|\frac{d^2y}{dx^2}\right|$, when $t = (\pi/2)$. MDF073
6. If $f(x) = mx + c$, $f(0) = f'(0) = 1$ then $f(2) =$ MDF021
7. $\frac{d}{dx} \tan^{-1}(\sec x + \tan x) = \lambda$ then $(10\lambda) = ?$ MDF022
8. $\frac{d}{dx} \left[2 \tan^{-1} \sqrt{\frac{1-\cos x}{1+\cos x}} \right] =$ MDF023
9. If $f(x) = \frac{1}{1-x}$, then the derivative of the composite function $f[f\{f(x)\}]$ is equal to MDF025
10. Let $y = 1 + \frac{x_1}{x-x_1} + \frac{x_2 \cdot x}{(x-x_1)(x-x_2)} + \frac{x_3 \cdot x^2}{(x-x_1)(x-x_2)(x-x_3)} + \dots$ upto $(n+1)$ terms
and if $\frac{x}{y} \frac{dy}{dx} = k \left[\frac{x_1}{x_1-x} + \frac{x_2}{x_2-x} + \frac{x_3}{x_3-x} + \dots + \frac{x_n}{x_n-x} \right]$, then k is MDF091
11. Let $x = \operatorname{cosec} \theta - \sin \theta$; $y = \operatorname{cosec}^6 \theta - \sin^6 \theta$ if $(x^2 + 4) \left(\frac{dy}{dx} \right)^2 - k(y^2 + 4) = 0$, then k is MDF092
12. If $x = \frac{1+t}{t^3}$, $y = \frac{3}{2t^2} + \frac{2}{t}$, then $x \left(\frac{dy}{dx} \right)^3 - \frac{dy}{dx}$ is equal to MDF030
13. If $y = ae^{mx} + be^{-mx}$, then $\frac{d^2y}{dx^2} - m^2y =$ MDF068

14. If $y = \tan^{-1} \frac{1}{x^2+x+1} + \tan^{-1} \frac{1}{x^2+3x+3} + \tan^{-1} \frac{1}{x^2+5x+7} + \tan^{-1} \frac{1}{x^2+7x+13} + \dots$ to 10 terms.
Find $(f'(0) + 1)^{-1}$.

MDF070

15. If $y = \cot^{-1} \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}$, find $2 \left| \frac{dy}{dx} \right|$ if $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$.

MDF071

16. A twice differentiable function $f(x)$ is defined for all real numbers and satisfies the following conditions

$$f(0) = 2; f'(0) = -5 \text{ and } f''(0) = 3.$$

The function $g(x)$ is defined by $g(x) = e^{ax} + f(x) \quad \forall x \in R$, where 'a' is any constant.

If $g'(0) + g''(0) = 0$, then number of possible value(s) of 'a' is

MDF072

17. Let $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3 \cdot (x^3 - y^3)$ and if $\frac{dy}{dx} = \frac{x^\ell}{y^\ell} \sqrt{\frac{1-y^k}{1-x^k}}$ (where $x \neq \pm 1$), then $\frac{k}{\ell}$ is

MDF093

18. If $f(2) = 4, f'(2) = 1$ then $\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x-2} =$

MDF026

EXERCISE - JEE (Main) PYQ

1. If $x = 3 \tan t$ and $y = 3 \sec t$, then the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$, is: [JEE (Main) 2019]

(1) $\frac{3}{2\sqrt{2}}$ (2) $\frac{1}{3\sqrt{2}}$ (3) $\frac{1}{6}$ (4) $\frac{1}{6\sqrt{2}}$

MDF035

2. Let $f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 \leq x \leq 2 \end{cases}$ and $g(x) = |f(x)| + f(|x|)$. Then, in the interval $(-2, 2)$, g is :-

[JEE (Main) 2019]

- (1) differentiable at all points (2) not differentiable at two points
(3) not continuous (4) not differentiable at one point

MDF036

3. For $x > 0$, if $(2x)^{2y} = 4e^{2x-2y}$, then $(1 + \log_e 2x)^2 \frac{dy}{dx}$ is equal to : [JEE (Main) 2019]

(1) $\log_e 2x$ (2) $\frac{x \log_e 2x + \log_e 2}{x}$ (3) $x \log_e 2x$ (4) $\frac{x \log_e 2x - \log_e 2}{x}$

MDF037

4. If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$ at $x = 0$ is equal to : [JEE (Main) 2019]

(1) $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$ (2) $\left(\frac{1}{e}, \frac{1}{e^2}\right)$ (3) $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$ (4) $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$

MDF038

5. The derivative of $\tan^{-1}\left(\frac{\sin x - \cos x}{\sin x + \cos x}\right)$, with respect to $\frac{x}{2}$, where $\left(x \in \left(0, \frac{\pi}{2}\right)\right)$ is :

[JEE (Main) 2019]

(1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) 1 (4) 2

MDF039

6. Let $y = y(x)$ be a function of x satisfying $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right) = -\frac{1}{4}$. Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$, is equal to : [JEE (Main) 2020]

(1) $\frac{\sqrt{5}}{2}$ (2) $-\frac{\sqrt{5}}{2}$ (3) $\frac{2}{\sqrt{5}}$ (4) $-\frac{\sqrt{5}}{4}$

MDF040

7. If $y(\alpha) = \sqrt{2\left(\frac{\tan \alpha + \cot \alpha}{1 + \tan^2 \alpha}\right) + \frac{1}{\sin^2 \alpha}}$, $\alpha \in \left(\frac{3\pi}{4}, \pi\right)$, then $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is : [JEE (Main) 2020]

(1) 4 (2) $-\frac{1}{4}$ (3) $\frac{4}{3}$ (4) -4

MDF041

8. If $y = \sum_{k=1}^6 k \cos^{-1} \left\{ \frac{3}{5} \cos kx - \frac{4}{5} \sin kx \right\}$, then $\frac{dy}{dx}$ at $x = 0$ is _____. [JEE (Main) 2020]

MDF042

9. Let $x^k + y^k = a^k$, ($a, K > 0$) and $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$, then k is : [JEE (Main) 2020]
- (1) $\frac{3}{2}$ (2) $\frac{1}{3}$ (3) $\frac{2}{3}$ (4) $\frac{4}{3}$

MDF094

10. If $(a + \sqrt{2} b \cos x)(a - \sqrt{2} b \cos y) = a^2 - b^2$, where $a > b > 0$, then $\frac{dx}{dy}$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ is : [JEE (Main) 2020]
- (1) $\frac{a-b}{a+b}$ (2) $\frac{a+b}{a-b}$ (3) $\frac{2a+b}{2a-b}$ (4) $\frac{a-2b}{a+2b}$

MDF095

11. Consider the function $f(x) = \begin{cases} \frac{P(x)}{\sin(x-2)} & ; x \neq 2 \\ 7 & ; x = 2 \end{cases}$

Where $P(x)$ is a polynomial such that $P''(x)$ is always a constant and $P(3) = 9$. If $f(x)$ is continuous at $x = 2$, then $P(5)$ is equal to _____. [JEE (Main) 2021]

MDF043

12. Let $f(x) = \cos \left(2 \tan^{-1} \sin \left(\cot^{-1} \sqrt{\frac{1-x}{x}} \right) \right)$, $0 < x < 1$. Then : [JEE (Main) 2021]
- (1) $(1-x)^2 f'(x) - 2(f(x))^2 = 0$ (2) $(1+x)^2 f'(x) + 2(f(x))^2 = 0$
 (3) $(1-x)^2 f'(x) + 2(f(x))^2 = 0$ (4) $(1+x)^2 f'(x) - 2(f(x))^2 = 0$

MDF044

13. If $y(x) = \cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right)$, $x \in \left(\frac{\pi}{2}, \pi \right)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{6}$ is : [JEE (Main) 2021]
- (1) $-\frac{1}{2}$ (2) -1 (3) $\frac{1}{2}$ (4) 0

MDF045

14. Let $f: S \rightarrow S$ where $S = (0, \infty)$ be a twice differentiable function such that $f(x+1) = xf(x)$. If $g: S \rightarrow R$ be defined as $g(x) = \log_e f(x)$, then the value of $|g''(5) - g''(1)|$ is equal to : [JEE (Main) 2021]

- (1) $\frac{205}{144}$ (2) $\frac{197}{144}$ (3) $\frac{187}{144}$ (4) 1

MDF096

15. If $y = y(x)$ is an implicit function of x such that $\log_e(x+y) = 4xy$, then $\frac{d^2y}{dx^2}$ at $x = 0$ is equal to. [JEE (Main) 2021]

MDF097

16. The value of $\log_e 2 \frac{d}{dx} (\log_{\cos x} \operatorname{cosec} x)$ at $x = \frac{\pi}{4}$ is [JEE (Main) 2022]
 (1) $-2\sqrt{2}$ (2) $2\sqrt{2}$ (3) -4 (4) 4

MDF046

17. Let $x(t) = 2\sqrt{2} \cos t \sqrt{\sin 2t}$ and $y(t) = 2\sqrt{2} \sin t \sqrt{\sin 2t}$, $t \in \left(0, \frac{\pi}{2}\right)$. Then $\frac{1 + \left(\frac{dy}{dx}\right)^2}{\frac{d^2 y}{dx^2}}$ at $t = \frac{\pi}{4}$ is equal to [JEE (Main) 2022]
 (1) $\frac{-2\sqrt{2}}{3}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $\frac{-2}{3}$

MDF047

18. If $y = \tan^{-1}(\sec x^3 - \tan x^3)$, $\frac{\pi}{2} < x^3 < \frac{3\pi}{2}$, then [JEE (Main) 2022]
 (1) $xy'' + 2y' = 0$ (2) $x^2 y'' - 6y + \frac{3\pi}{2} = 0$
 (3) $x^2 y'' - 6y + 3\pi = 0$ (4) $xy'' - 4y' = 0$

MDF048

19. Let $f(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, $a \in R$. Then the sum of the squares of all the values of a for which $2f'(10) - f'(5) + 100 = 0$ is : [JEE (Main) 2022]
 (1) 117 (2) 106 (3) 125 (4) 136

MDF049

20. If $y(x) = (x^x)^x$, $x > 0$ then $\frac{d^2 x}{dy^2} + 20$ at $x = 1$ is equal to: [JEE (Main) 2022]

MDF050

21. If $y(x) = x^x$, $x > 0$, then $y''(2) - 2y'(2)$ is equal to : [JEE (Main) 2023]
 (1) $8 \log_e 2 - 2$ (2) $4 \log_e 2 + 2$ (3) $4 (\log_e 2)^2 - 2$ (4) $4 (\log_e 2)^2 + 2$

MDF051

22. Let $y = f(x) = \sin^3 \left(\frac{\pi}{3} \left(\cos \left(\frac{\pi}{3\sqrt{2}} (-4x^3 + 5x^2 + 1)^{\frac{3}{2}} \right) \right) \right)$. Then, at $x = 1$, [JEE (Main) 2023]
 (1) $2y' + \sqrt{3}\pi^2 y = 0$ (2) $2y' + 3\pi^2 y = 0$ (3) $\sqrt{2}y' - 3\pi^2 y = 0$ (4) $y' + 3\pi^2 y = 0$

MDF052

23. Let $y(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$. Then $y' - y''$ at $x = -1$ is equal to [JEE (Main) 2023]
 (1) 976 (2) 464 (3) 496 (4) 944

MDF053

24. Let $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}$, $x \in [0, \pi] - \left\{\frac{\pi}{4}\right\}$. Then $f\left(\frac{7\pi}{12}\right) f''\left(\frac{7\pi}{12}\right)$ is equal to [JEE (Main) 2023]

- (1) $\frac{-2}{3}$ (2) $\frac{2}{9}$ (3) $-\frac{1}{3\sqrt{3}}$ (4) $\frac{-2}{3\sqrt{3}}$

MDF054

25. If $2x^y + 3y^x = 20$, then $\frac{dy}{dx}$ at $(2, 2)$ is equal to [JEE (Main) 2023]

- (1) $-\left(\frac{3 + \log_e 8}{2 + \log_e 4}\right)$ (2) $-\left(\frac{2 + \log_e 8}{3 + \log_e 4}\right)$ (3) $-\left(\frac{3 + \log_e 16}{4 + \log_e 8}\right)$ (4) $-\left(\frac{3 + \log_e 4}{2 + \log_e 8}\right)$

MDF055

26. If $y(\theta) = \frac{2\cos\theta + \cos 2\theta}{\cos 3\theta + 4\cos 2\theta + 5\cos\theta + 2}$, then at $\theta = \frac{\pi}{2}$, $y'' + y' + y$ is equal to: [JEE (Main) 2024]

- (1) $\frac{3}{2}$ (2) 1 (3) $\frac{1}{2}$ (4) 2

MDF148

27. Let $f(x) = x^5 + 2x^3 + 3x + 1$, $x \in R$, and $g(x)$ be a function such that $g(f(x)) = x$ for all $x \in R$. Then $\frac{g(7)}{g'(7)}$ is equal to: [JEE (Main) 2024]

- (1) 7 (2) 42 (3) 1 (4) 14

MDF149

28. If $y = \frac{(\sqrt{x} + 1)(x^2 - \sqrt{x})}{x\sqrt{x} + x + \sqrt{x}} + \frac{1}{15}(3\cos^2 x - 5)\cos^3 x$, then $96y'\left(\frac{\pi}{6}\right)$ is equal to: [JEE (Main) 2024]

MDF150

29. Let for a differentiable function $f: (0, \infty) \rightarrow R$, $f(x) - f(y) \geq \log_e \left(\frac{x}{y}\right) + x - y$, $\forall x, y \in (0, \infty)$. Then

$\sum_{n=1}^{20} f'\left(\frac{1}{n^2}\right)$ is equal to ____.

[JEE (Main) 2024]

MDF151

30. Suppose

$$f(x) = \frac{(2^x + 2^{-x}) \tan x \sqrt{\tan^{-1}(x^2 - x + 1)}}{(7x^2 + 3x + 1)^3},$$

Then the value of $f'(0)$ is equal to

[JEE (Main) 2024]

- (1) π (2) 0 (3) $\sqrt{\pi}$ (4) $\frac{\pi}{2}$

MDF152

EXERCISE - JEE (Advanced) PYQ

1. Let $g(x) = \ln f(x)$ where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = xf(x)$. Then for $N = 1, 2, 3, \dots$, $g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$ [JEE (Advanced) 2008]

(A) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$ (B) $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$
 (C) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$ (D) $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$

MDF074

2. Let f and g be real valued functions defined on interval $(-1, 1)$ such that $g''(x)$ is continuous, $g(0) \neq 0$, $g'(0) = 0$, $g''(0) \neq 0$, and $f(x) = g(x) \sin x$.

Statement-1 : $\lim_{x \rightarrow 0} [g(x) \cot x - g(0) \operatorname{cosec} x] = f''(0)$

and

Statement-2 : $f'(0) = g(0)$

[JEE (Advanced) 2008]

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation of statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

MDF075

3. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is

[JEE (Advanced) 2009]

MDF076

4. Let $f(\theta) = \sin \left(\tan^{-1} \left(\frac{\sin \theta}{\sqrt{\cos 2\theta}} \right) \right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Then the value of $\frac{d}{d(\tan \theta)} (f(\theta))$ is

[JEE (Advanced) 2011]

MDF077

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then-

[JEE (Advanced) 2016]

(A) $g'(2) = \frac{1}{15}$ (B) $h'(1) = 666$ (C) $h(0) = 16$ (D) $h(g(3)) = 36$

MDF079

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	A	B	B	B	A	A	B	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A	C	B	D	B	C	B	A
Que.	21	22	23							
Ans.	C	C	D							

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,D	A,B	B,D	A,B,C,D	B,C,D	A,B,C,D	A,B	A,D	A	B
Que.	11	12	13	14	15	16				
Ans.	C	A	B	D	C	C				

EXERCISE (S)

1.	0	2.	0	3.	1	4.	2	5.	3.00
6.	3	7.	5	8.	1	9.	1	10.	1.00
11.	36.00	12.	1	13.	0	14.	101	15.	1.00
16.	2	17.	3.00	18.	2				

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	4	1	4	2	1	91	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	39	3	1	1	40	4	4	2	3	16
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	3	2	2	4	4	105	2890	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5					
Ans.	A	A	2	1	B,C					

Important Notes